Yingrui Hou

Simulation & Model Validation | Digital Twin Applications

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience connecting measurements to simulation, calibrating models and diagnosing discrepancies. Built Python/C++/Geant4 workflows; learned TensorFlow through code adaptation and simulation-based verification, then migrated computation to a shared GPU workstation. Brings research coordination and technical communication to simulation engineering and digital twin opportunities.

Core skills

- Simulation: Geant4/C++, detector geometry, interaction stepping, response logic, event-level outputs and measured response maps.
- Validation: simulation-to-measurement alignment, deep-learning calibration, image-recognition model checks, bias correction and uncertainty analysis.
- Signals & models: time-profile decomposition, likelihood inference; CatBoost five-fold assessment and train/test KS checks, control-sample reweighting and threshold optimisation.
- Computing: Python, C/C++, ROOT/RooFit, NumPy, pandas; TensorFlow/GPU code adaptation and verification; Linux and shared GPU workstation setup.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Developed a Shell pipeline for beam-test acquisition, storage, quick calibration and initial quality checks, connecting detector readout, tracking and timing calibration. Participated in on-site setup and data taking; assisted subsequent fitting and measurement-effect corrections.
- Connected beam-test measurements and position-dependent response maps to simulated energy deposits, evaluating performance against measured rather than ideal response.
- Developed time-resolved signal analysis and ROOT/C++ correction workflows; checked reconstructed energy before and after correction under the studied conditions.
- Diagnosed incorrect ECAL photon truth associations; revised cluster-energy matching, tested multiple simulated decay channels and improved thread safety through review. The merged change supported subsequent Run 3 analyses and reconstruction-performance checks.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Measurement-informed detector model

Developed Shell automation linking beam-test acquisition, storage, rapid calibration and quality checks. Participated in setup and data taking, and assisted later analysis. Connected measured response maps with Geant4 simulation; non-uniformity contributed below 1% to the constant term in the first prototype study.

Signal decomposition and correction

Extended scintillation simulations and fitted signal fractions for energy correction. The team chose $ZnWO_4$ after measured BGO profiles showed insufficient low-energy separation from electromagnetic excitation. Separately, a representative rotation-based simulation study reduced a resolution term by about 2×; results apply to that configuration.

ECAL diagnosis and integrated software correction

Traced anomalous simulated photon resolution to incorrect truth associations under high occupancy. Revised matching using normalised truth-particle energy deposits within the photon cluster. Checked identities, energies and event histories across several simulated decay channels. Improved thread safety following review; the MR was merged and used in Run 3 analyses and reconstruction-performance checks.

Learning TensorFlow and delivering GPU computation

Learned TensorFlow from inherited code; personally ported amplitude models, added efficiency, resolution and bias effects, and validated calculations against mature C++ code at identical sampling points. Resolved discrepancies by aligning variable types; checked simulated observables against real data with KS tests. Set up a shared GPU workstation in half a day, then achieved a GPU run in another half-day. Contributed to testing a collaborator-developed fitter.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Learning, delivery & collaboration

- Learning by implementation: extended unfamiliar TensorFlow code with new research models and checked it on generated data; applied CatBoost and Geant4 in other projects.
- Delivery: agreed component interfaces and variable definitions before coding; reassessed priorities and timelines with collaborators as data processing exposed new issues.
- Collaboration: GitLab development and mutual merge-request review; identified a fitter sign error through mathematical checks. Weekly planning and rapid technical discussions; research presentations at GDR-InF 2022 and FCC/DRD France 2025.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Measurement-informed detector model (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)
- Signal decomposition and correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-5)
- ECAL diagnosis and integrated software correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Learning TensorFlow and delivering GPU computation (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)